

INTRINSIC METRIC DATA AND CANCELLATION LAWS FOR POLYNOMIAL OBSERVATIONS

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ABSTRACT. A polynomial observation germ carries metric information which is not determined by its contact orders. We identify a canonical closed convex set of residual tensors whose positive definite support data are exactly all leading entrance coefficients. For full-rank binary stochastic polynomial observations, its constrained local sections are quadratic quotient forms. We characterize precisely the finite ray probes which determine such a form, prove an adaptive query lower bound over the class of observation metrics, and identify the smaller piecewise-quadratic invariant seen after one-sided endpoint variables are eliminated. For polynomial probability curves we obtain a relative distance factorization, uniform through arbitrary cancellations of a finite residual polynomial. It gives the exact identification wall, every opening along a bounded semialgebraic parameter arc, and the crossover coefficient. The real divisorial order formula computes the exponent; the tensor set retains the metric information which that formula discards. The complete multiplicity theorem, rank-deficient cubic classification, two cubic walls, weighted real comparisons, and their proofs accompany the principal article without deletion.

1. INTRODUCTION

Two different questions arise in a polynomial inverse problem. Near an identifying observation, which joint root displacements are compatible with a small observation error? Near a nonidentifying observation, at what distance does a remote representative enter the observation ball? Orders alone answer neither question sharply. The first also requires a quotient metric on the admissible root directions. The second requires the leading residual vectors, and not only their orders of vanishing.

We use the same construction for both questions. Given a set E of leading residual vectors, form

$$\mathcal{C}(E) = \text{cl}(\text{conv}\{yy^\top : y \in E\} + \text{Sym}_M^+). \quad (1.1)$$

Here Sym_M^+ is the cone of positive semidefinite real symmetric matrices. For every positive definite observation metric H ,

$$\inf_{S \in \mathcal{C}(E)} \text{tr}(HS) = \inf_{y \in E} y^\top Hy.$$

Conversely, these infima, for all $H > 0$, determine $\mathcal{C}(E)$. The substantive issue is to identify the correct leading residual set and to prove that minimization in the original experiment converges to minimization over that set.

Theorem 2.2 solves this issue for an isolated real semialgebraic remote germ. It yields a canonical pair $(\rho, \mathcal{C})$: the exponent is independent of the positive definite norm, and $\mathcal{C}$ is a complete invariant of the leading coefficient as the norm varies. A real monomial presentation computes the pair but is not part of its definition. In particular, no choice of exceptional divisors is declared canonical.

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For the binary stochastic polynomial model, the constrained local leading residuals are affine spaces $La + Dv$. Their metric minima are quadratic forms $v^\top Qv$. Theorem 3.3 characterizes all sufficient finite ray-probe families. The familiar polarization family is optimal for arbitrary positive definite observation metrics, even against deterministic adaptive exact queries. This is an oracle lower bound, not a sampling theorem. For a fixed Hellinger family, the exact condition is a secant-kernel criterion; the larger metric-class lower bound is not silently transferred to that smaller family. One-sided endpoint directions require an additional distinction: the leading spectrum can see only a piecewise-quadratic partial minimum, not the entire matrix. Theorem 3.5 identifies this effective invariant.

Theorem 4.1 is the principal concrete consequence for remote identification. For

$$(x^m, g_a(x)), \quad g_a(x) = \sum_{i=1}^s a_i x^{n_i}, \quad m < n_1 < \cdots < n_s,$$

observed near $(\delta, 0)$, put $z = \delta^{1/m}$. The distance is

$$d_H(a, \delta) = \kappa_H |g_a(z)| (1 + O(z^{n_1-m})), \quad \kappa_H^2 = h_{22} - h_{12}^2/h_{11}.$$

The assertion is a two-sided relative estimate, including the exact zero set, uniformly for bounded coefficients. It therefore remains informative on moving parameter paths where several leading monomials cancel. It classifies the exact wall $g_a(z) = 0$, all nonzero arc openings, and the transition between successive contact orders. A strictly positive polynomial probability realization gives the corresponding Hellinger statement with its Fisher-normalized coefficient.

Relation to the classical order theory. Integral closure, normalized blow-ups, and divisorial order tests already provide the classical mechanism for rational contact exponents; the complex-analytic account and real-analytic appendix in [1] are directly relevant. We do not claim a new valuative criterion for integral closure. Our real sectors encode the feasible inequalities and positive time before orders are taken. Beyond the order, we retain the residual tensor envelope and prove the cancellation-uniform wall formula. The latter is not a consequence of knowing a single exponent. Likewise, the signed real tropical framework of [2] records signs and valuations, whereas cancellation of equal-weight terms depends on their residues. Corollary 4.4 specifies exactly how that distinction controls the wall in this family.

Organization and the accompanying mathematical record. Section 2 constructs the canonical metric invariant. Section 3 develops the binary local theory, observable-law formulation, exact probe criterion, and endpoint elimination. Section 4 proves the uniform polynomial wall classification. Appendix A specifies the real monomial presentation and relates it to the intrinsic invariant. Appendix B records the exact relation to the accompanying results.

The supporting volume is the complete preceding article, not an abridgment. It contains the full root-multiplicity and boundary atlas; exact rank-deficient cubic fibre; inverse through channel-rank loss; weight and endpoint walls; signed second-order and equality-shell computations; real weighted atlas; Puiseux-residue and finite-jet comparisons; separation results; and fixed-format entrance theory. Their original hypotheses and conclusions remain in force. This principal article reorganizes the new argument rather than repeating that entire sequence as its introduction.

2. THE INTRINSIC METRIC CONTACT ENVELOPE

Let N be a compact semialgebraic set, let $R : N \times [0, \delta_0] \rightarrow \mathbb{R}^M$ be continuous and semialgebraic, and suppose that $R(x, 0) = 0$ has just one solution x_* in N . We assume

$$\min_{x \in N} |R(x, \delta)| > 0 \quad (0 < \delta \ll 1), \quad R(x_*, \delta) \longrightarrow 0. \quad (2.1)$$

All residual coordinates and the positive time parameter are fixed. For $H > 0$ set

$$d_H(\delta) = \min_{x \in N} \|R(x, \delta)\|_H, \quad \|y\|_H^2 = y^\top H y.$$

Compact attainment and semialgebraicity imply a positive leading power for d_I . Write

$$d_I(\delta) = a_I \delta^\rho + o(\delta^\rho), \quad a_I > 0, \quad \rho \in \mathbb{Q}_{>0}. \quad (2.2)$$

This elementary one-variable consequence of real curve parametrization is also proved, uniformly on strata, in the accompanying entrance theorem.

Definition 2.1. The leading residual set is

$$\mathcal{E}_\rho = \{y \in \mathbb{R}^M : (0, y) \in \text{cl}\{(\delta, \delta^{-\rho} R(x, \delta)) : x \in N, 0 < \delta < \delta_0\}\}.$$

The metric contact envelope is $\mathcal{C}_\rho = \mathcal{C}(\mathcal{E}_\rho)$ as in (1.1).

The set $\mathcal{E}_\rho$ need not be bounded, convex, or a cone. Taking its convex tensor envelope is a deliberate quotient: the resulting invariant records all positive definite metric minima, not every leading residual vector.

Theorem 2.2 (Complete metric contact data). *Under (2.1), $\mathcal{E}_\rho$ is a nonempty closed semialgebraic set and $\text{dist}(0, \mathcal{E}_\rho) = a_I > 0$. For every $H > 0$,*

$$d_H(\delta) = a_H \delta^\rho + o(\delta^\rho), \quad a_H^2 = \min_{y \in \mathcal{E}_\rho} y^\top H y = \inf_{S \in \mathcal{C}_\rho} \text{tr}(HS) > 0. \quad (2.3)$$

The exponent is the same for all $H > 0$. Moreover,

$$\mathcal{C}_\rho = \bigcap_{H > 0} \{S \in \text{Sym}_M : \text{tr}(HS) \geq a_H^2\}. \quad (2.4)$$

Consequently two such germs have the same leading coefficient for every positive definite metric if and only if their metric contact envelopes agree, after matching their residual coordinates and exponents.

A linear change of residual coordinates $R' = TR$, $T \in \text{GL}_M(\mathbb{R})$, sends $\mathcal{E}_\rho$ to $T\mathcal{E}_\rho$ and $\mathcal{C}_\rho$ to $TC_\rho T^\top$. A surjective reparametrization of the feasible residual germ leaves these objects unchanged. The function $H \mapsto a_H^2$ is positive homogeneous, concave, monotone in the positive semidefinite order, and locally Lipschitz on Sym_M^{++} .

Proof. The defining closure is semialgebraic and closed, so its zero-time section is closed and semialgebraic. Choose minimizers for d_I . Their normalized residuals are bounded by (2.2); a convergent subsequence yields a point of $\mathcal{E}_\rho$ of norm a_I . Conversely, along every sequence producing a point $y \in \mathcal{E}_\rho$, the inequality $|R(x, \delta)| \geq d_I(\delta)$ gives $|y| \geq a_I$. This proves the first assertions.

Norm equivalence gives constants $0 < c < C$ with $cd_I \leq d_H \leq Cd_I$. The positive semialgebraic function d_H has a leading power; these two inequalities force its exponent to be ρ and its coefficient to be positive. Normalized residuals of its minimizers are bounded, again by norm equivalence. A convergent subsequence has limit $y_H \in \mathcal{E}_\rho$ and $\|y_H\|_H = a_H$. Every other point of $\mathcal{E}_\rho$ is obtained along a sequence, on which $d_H \leq \|R\|_H$ gives $a_H \leq \|y\|_H$. Thus the first minimum in (2.3) is attained. A linear functional bounded below on a set has the same infimum on its convex hull and closure; adding Sym_M^+ cannot lower $\text{tr}(HS)$ when $H > 0$. This proves the second equality.

For completeness, we prove that positive definite functionals suffice in (2.4). The inclusion from left to right is immediate. If $S_0 \notin \mathcal{C}_\rho$, strong separation of a point from a closed convex set gives a nonzero symmetric G with

$$\text{tr}(GS_0) < \inf_{S \in \mathcal{C}_\rho} \text{tr}(GS).$$

Since $\mathcal{C}_\rho + \text{Sym}_M^+ = \mathcal{C}_\rho$, finiteness of the infimum forces $G \geq 0$: otherwise adding a large multiple of $vv^\top$ with $v^\top G v < 0$ would send it to $-\infty$. Also $\mathcal{C}_\rho \subseteq \text{Sym}_M^+$. For sufficiently small $\varepsilon > 0$ the strict separation persists for $G + \varepsilon I$, because its infimum on $\mathcal{C}_\rho$ is at least the old infimum whereas its value at S_0 tends to the old value. Now $G + \varepsilon I > 0$, proving the reverse inclusion.

Coordinate covariance follows by applying T to the normalized residuals and observing that congruence maps the positive semidefinite cone onto itself. The reparametrization assertion follows from equality of the feasible residual images as germs. The infimum of linear functions of H is concave and positive homogeneous, and each $yy^\top$ is positive semidefinite, giving monotonicity. On a compact metric set $\lambda I \leq H \leq \Lambda I$, fix $y_0 \in \mathcal{E}_\rho$. Every minimizer satisfies $\lambda|y_H|^2 \leq \Lambda|y_0|^2$. Testing the minimum for H' at y_H and conversely gives

$$|a_H^2 - a_{H'}^2| \leq (\Lambda/\lambda)|y_0|^2 \|H - H'\|_{\text{op}}.$$

This proves local Lipschitz continuity. $\square$

Remark 2.3 (What is canonical). The pair $(\rho, \mathcal{C}_\rho)$ is defined in the original real residual space. It depends on the feasible germ and the positive time coordinate. It is not an invariant of an unmarked complex ideal alone. Its first entry records the optimal real contact order. Its second entry is exactly the information needed to vary the leading observation metric. It does not purport to recover nonconvex residual geometry invisible to all such minima.

Proposition 2.4 (Arc form and the optimal exponent). *For R as above,*

$$\rho = \sup_{\gamma} \frac{\min_i \text{ord}(R_i \circ \gamma)}{\text{ord}(\delta \circ \gamma)}, \quad (2.5)$$

where γ ranges over positive-time feasible semialgebraic Puiseux arcs tending to $(x_*, 0)$. The supremum is attained. It is also the least real c for which $\delta^c \leq C|R(x, \delta)|$ throughout the feasible germ, for some $C > 0$.

Proof. The lower estimate $|R| \geq (a_I/2)\delta^\rho$ bounds every arc ratio above by ρ and gives the inequality at $c = \rho$. Semialgebraic choice selects minimizers of d_I as a function of positive time; after shortening the interval they form a Puiseux arc. Their source points tend to x_* , since any other limiting point would be another zero of $R(\cdot, 0)$. Along this arc $|R|$ has order ρ in time. No $c < \rho$ can satisfy the proposed bound there. For a real vector of Puiseux germs, the norm order is the minimum of the coordinate orders, because sums of squares cannot cancel. This proves (2.5). $\square$

3. LOCAL INFORMATION IN BINARY POLYNOMIAL OBSERVATIONS

We now identify the constrained version of the preceding metric construction in an observation model with a spectral target. Fix $D > 0$, $d \geq 1$, and $0 < \pi < 1/2$. Let f_1, f_2 be monic degree- d polynomials with real roots in $[0, D]$. Let $\alpha_1 + \alpha_2 = 1$, $\alpha_a \geq \pi$, and let $U = (u_1, u_2)$ and $V = (v_1, v_2)$ be nonnegative column-stochastic 2×2 matrices. Put

$$K(z) = \sum_{a=1}^2 \alpha_a f_a(z) u_a v_a^\top, \quad q(z) = \mathbf{1}^\top K(z) \mathbf{1}, \quad P_j = K(T_j)/q(T_j), \quad (3.1)$$

where $D < T_1 < \dots < T_\ell$ and $\ell \geq 2d + 1$. The target is the unordered root multiset of $f_1 f_2$, with its bottleneck metric. The observation distance is

$$h_w(P, P')^2 = \sum_j w_j \sum_e (\sqrt{P_{je}} - \sqrt{P'_{je}})^2, \quad w_j > 0, \quad \sum_j w_j = 1.$$

A centre is called full rank here if f_1, f_2 are coprime, U, V are invertible and strictly positive, and both weights are strictly larger than π . Competitors remain in the entire closed model.

Lemma 3.1 (The coefficient inverse). *At a full-rank centre the observation determines the two component polynomials, weights and channels up to a common component permutation. Near its observation the recovered coefficient and stochastic coordinates are Lipschitz, with constants uniform on compact sets of full-rank centres. Their derivative in unconstrained coefficient and stochastic coordinates has a left inverse.*

Proof. First recover the monic normalizer. A homogeneous ambiguity consists of a polynomial e of degree below d and a matrix polynomial H of degree at most d with $H(T_j) = e(T_j)P_j$. The degree- $2d$ polynomial $qH - eK$ vanishes at $2d + 1$ clocks, so it vanishes identically. Applying the invertible second channel to the second marginal gives $q \mid ef_1$ and $q \mid ef_2$. Coprimality implies $q \mid e$, hence $e = 0$ and $H = 0$. Thus the finite linear system for the monic q and K has full column rank. A fixed nonsingular minor supplies an analytic local recovery; subtraction of the equations and the compact coefficient bounds of the closed model give the Lipschitz estimate for all close competitors.

There are two clocks T_i, T_j at which $f_1(T_j)/f_1(T_i) \neq f_2(T_j)/f_2(T_i)$; otherwise a nonzero polynomial of degree at most d would vanish at all the clocks. The matrix $K(T_j)K(T_i)^{-1}$ has these simple eigenvalues and eigenvectors u_1, u_2 . Normalize the eigenvectors by their entry sums. The row sums of $U^{-1}K(z)$ then recover $\alpha_a f_a(z)$; their leading coefficients recover α_a , and their normalized rows recover V . These operations are analytic near the centre. They also apply to every exact competitor and prove uniqueness. A finite cover gives the compact-uniform estimates. Differentiating the analytic recovery gives the asserted left inverse. $\square$

3.1. Observable-law coordinates and their variational costs. Fix a full-rank centre observation p . Mark the distinct clusters by their recovered component, root value, multiplicity, and endpoint status. Within a sufficiently small observation neighbourhood this marking is determined by p , up to the common component permutation. It is not an extra unknown latent label. Multiplicities can be specified by polynomial gcd conditions and endpoint evaluation. They need not be continuous across a change of stratum.

For nearby laws in the model image, isolate the roots near each central root r_j , of multiplicity m_j . Their monic cluster factor is an analytic function of the recovered coefficients. Writing the root displacements as $u_{j,i}$, the quantities

$$s_j = \sum_i u_{j,i}, \quad X_j = \frac{1}{2} \sum_i (u_{j,i} - s_j/m_j)^2$$

are analytic functions of that factor's first two coefficients. Individual repeated roots need not be differentiable. At D reverse the sign of s_j ; at zero keep its sign.

If an interior root is repeated, retain its X_j and retain the signed sums at every endpoint cluster. All interior sums and stochastic variables are free. The retained cone is $\mathcal{K} = \mathbb{R}_+^k$ and the root scale is $t^{1/2}$. If no interior root is repeated, retain all cluster sums; then $\mathcal{K} = \mathbb{R}^{k_0} \times \mathbb{R}_+^{k_1}$, only stochastic variables are free, and the root scale is t . In either case $k \leq 2d$. Let τ_p be the vector of these retained *functions on the observable model image*, centred at p .

For $v \in \mathcal{K}$, define

$$c_p(v) = \lim_{t \downarrow 0} \frac{4}{t^2} \inf \{ h_w(p', p)^2 : p' \text{ is a nearby model law, } \tau_p(p') = tv \}. \quad (3.2)$$

This is a model-dependent variational oracle. It is neither a single empirical statistic nor a claim that latent root constraints can be physically imposed. Its law-space definition removes a choice of nuisance parametrization; it does not remove the model from the definition. Since the centre is identifying, its pattern and the oracle are computable from the exact centre law and the known model. This observation by itself is not a compression theorem.

Proposition 3.2 (The local quotient and the leading root set). *The limits (3.2) exist. If $La + Dv$ is the first observation score in free and retained variables, and $W = \text{diag}(w_j/p_{je})$, then*

$$c_p(v) = v^\top Q v, \quad Q = D^\top W D - D^\top W L (L^\top W L)^{-1} L^\top W D > 0. \quad (3.3)$$

An empty free block contributes zero. These costs are intrinsic to the observable germ with its spectral coordinate and marking. Reparametrizing free directions or adding a free direction to a retained score column leaves them unchanged.

The complete leading root set is the decoding of $\{v \in \mathcal{K} : v^\top Q v \leq 4\}$. In the repeated regime, decode each retained variance v_j into all ordered zero-sum vectors y_j with $\frac{1}{2} \sum_i y_{j,i}^2 = v_j$, and put zero in the other root blocks; endpoint sums are existentially minimized. In the linear regime decode sums into the simple interior increments and all ordered one-sided endpoint increments with those sums. The scaled whole-model root sets converge to these sets in Hausdorff distance, with error $O(\sqrt{t})$ and $O(t)$ respectively, uniformly on compact subsets of a fixed pattern stratum.

Proof. The first coefficient increment in component a is

$$-\sum_{j \in a} s_j \frac{f_a(z)}{z - r_j} - \sum_{\substack{j \in a \text{ repeated} \\ 0 < r_j < D}} X_j \frac{f_a(z)}{(z - r_j)^2}.$$

The simple and double partial fractions at distinct poles are linearly independent. Lemma 3.1 therefore makes the combined score (L, D) injective. Orthogonal projection off $\text{ran } L$ proves (3.3) and positive definiteness.

For the passage from the actual law-space constraint to this minimum, choose at each repeated interior cluster a zero-sum vector y of variance v_j , and use roots $r_j + ta_j/m_j + \sqrt{t} y_i$. Every nonnegative variance has such a vector; one may take $m_j - 1$ equal entries and their negative sum. At endpoints use equal one-sided displacements with the prescribed sum. Give every stochastic variable its free order- t increment. Strict margins at the centre ensure feasibility. These competitors have retained coordinates exactly tv and satisfy

$$F(\xi) - p = t(La + Dv) + O(t^{3/2}), \quad 4h_w(F(\xi), p)^2/t^2 = \|La + Dv\|_W^2 + O(\sqrt{t}).$$

In the linear regime the errors are $O(t^2)$ and $O(t)$. The minimizing free vector is bounded on compact retained sets, giving the upper bound.

Conversely an almost-minimizer has observation error $O(t)$ and therefore coefficient and stochastic errors $O(t)$. Analytic cluster factorization gives cluster coefficient errors of this order. The first coefficient bounds all sums. The real identity for the second centred coefficient, $e_2 = -\frac{1}{2} \sum_i (u_i - s/m)^2$, bounds interior centred displacements by $O(\sqrt{t})$. At endpoints every displacement has the same sign and is bounded by the sum, hence is $O(t)$. Thus the same expansions hold for all almost-minimizers, with bounded free variables, and give the lower bound. Compactness allows minima after restricting to a small closed cluster neighbourhood; the coefficient inverse forces the small-cost minimizers into every smaller such neighbourhood. The limit is independent of that restriction and is the law-space cost in (3.2).

The same bounds apply to all competitors in the observation ball. Divide the root blocks by the stated scale and apply the score expansion. A radial contraction of the retained and free order- t variables by $1 - O(\sqrt{t})$, or $1 - O(t)$ in the linear regime, puts their limiting score in the norm ball of radius two. Repeated root vectors contract by the square root of that factor. This proves the outer Hausdorff bound. For the reverse bound use the feasible root constructions above with the same contraction, choosing its constant to dominate the Taylor remainder. Both estimates are uniform on compact pattern sets by the uniform inverse and strict feasibility margins. Exact identification excludes other limiting components of the whole-model ball. Cluster separation makes the bottleneck metric the maximum of the ordered within-cluster coordinate metrics. This

proves the assertion for the joint set, not merely its diameter. The invariance claims follow from the quotient by $\text{ran } L$ or directly from (3.2). $\square$

The affine residual set $E_v = \{La + Dv : a \in \mathbb{R}^r\}$ has a tensor envelope as in (1.1). Its metric support at W is exactly $c_p(v)$. Thus the local quotient form is a constrained section of the same construction used in Theorem 2.2; no remote order argument is needed to prove the local inverse.

3.2. Which finite probes are sufficient? Let $k \geq 1$, let $\mathcal{K} \subseteq \mathbb{R}^k$ be either cone just described, and let $v_1, \dots, v_m \in \mathcal{K}$. For $H \in \text{Sym}_k$ define the linear sampling map

$$\mathcal{A}_V(H) = (v_1^\top H v_1, \dots, v_m^\top H v_m).$$

For a family $\mathcal{Q} \subseteq \text{Sym}_k^{++}$, its secant set is $\mathcal{Q} - \mathcal{Q} = \{Q - Q' : Q, Q' \in \mathcal{Q}\}$.

Theorem 3.3 (Exact probe criterion and adaptive lower bound). *The ray probes determine Q on $\mathcal{Q}$ if and only if*

$$\ker \mathcal{A}_V \cap (\mathcal{Q} - \mathcal{Q}) = \{0\}. \quad (3.4)$$

On all positive definite forms this is equivalent to

$$\text{span}\{v_i v_i^\top : 1 \leq i \leq m\} = \text{Sym}_k. \quad (3.5)$$

In particular at least $k(k+1)/2$ ray values are required. The legal probes e_i and $e_i + e_j$, $i < j$, attain this bound. No deterministic adaptive algorithm making fewer exact ray queries in the worst case can recover every positive definite form.

The same lower bound applies to recovery of the complete leading spectral fibre on an endpoint-free pattern, when the class of observation metrics is allowed to vary over all positive definite forms on the observation tangent space. It is not asserted for the fixed Hellinger metric family, for which (3.4) is the exact statement. When endpoint variables are invisible at the leading root scale, the smaller invariant in Theorem 3.5 replaces the full form.

Proof. Two forms produce the same values precisely when their difference is in $\ker \mathcal{A}_V$, proving (3.4). If (3.5) fails, choose a nonzero symmetric H orthogonal to all $v_i v_i^\top$. For any $Q_0 > 0$, both $Q_0 + \varepsilon H$ and $Q_0 - \varepsilon H$ are positive definite for sufficiently small $\varepsilon > 0$, and have the same values. Conversely (3.5) makes the linear map injective. Dimension gives the lower bound. For the specified family,

$$Q_{ii} = c(e_i), \quad Q_{ij} = \frac{1}{2}(c(e_i + e_j) - c(e_i) - c(e_j)),$$

so the bound is attained using vectors in either feasible cone.

For the adaptive statement, run the proposed deterministic algorithm at $Q_0 > 0$ and record its chosen directions. If their number is less than $\dim \text{Sym}_k$, choose $H \neq 0$ orthogonal to every recorded tensor. The two nearby forms $Q_0 \pm \varepsilon H$ give exactly the recorded answers. Induction on the query number shows that the algorithm chooses the same next direction, and makes the same stopping decision, on both inputs. It must therefore output the same answer for two different forms. This argument does not assume continuity of the rule choosing queries.

It remains to justify the observation-model and spectral qualifications. At a fixed binary germ the combined score (L, D) is injective. On its image choose an inner product which makes the free and retained score subspaces orthogonal and gives the retained subspace any prescribed Gram matrix $Q > 0$. Extend it positively to an ambient complement. A constant ambient metric already realizes this tangent metric. Profiling then gives precisely the prescribed Q . This proves realizability over the declared metric class, not over the smaller family obtained by changing only Hellinger design weights.

On an endpoint-free pattern every retained variable is visible in the leading root vector: it is a cluster variance in the repeated regime and an individual simple-root displacement in the linear regime. All v in the retained cone have a decoded root vector. Thus the leading fibre determines

the cone section $\{v \in \mathcal{K} : v^\top Q v \leq 4\}$. Its radial function determines $v^\top Q v$ for all nonzero $v \in \mathcal{K}$. Since $\mathcal{K}$ has nonempty interior, equality there forces equality of the two quadratic polynomials. Two distinct forms in the preceding adversary argument therefore give distinct spectral fibres. This proves the fibre lower bound with its stated restriction. $\square$

Corollary 3.4 (Stability of a sufficient probe family). *Suppose $\mathcal{A}_V$ is injective and let $\sigma_V > 0$ be its least singular value for the Frobenius and Euclidean norms. Least-squares reconstruction from an error vector of norm at most ε gives*

$$\|\hat{Q} - Q\|_{\text{op}} \leq \|\hat{Q} - Q\|_F \leq \varepsilon / \sigma_V.$$

If $Q \geq \lambda I$ and $\eta = \varepsilon / \sigma_V < \lambda$, put $\vartheta = \eta / \lambda$. For a fixed decoder of degree $\gamma \in \{1/2, 1\}$, the reconstructed leading sets satisfy

$$(1 + \vartheta)^{-\gamma/2} \mathcal{L}(Q) \subseteq \mathcal{L}(\hat{Q}) \subseteq (1 - \vartheta)^{-\gamma/2} \mathcal{L}(Q).$$

Their diameters obey the same multiplicative bounds.

Proof. The left inverse of the sampling map has norm $1/\sigma_V$. The form bounds $(1 - \vartheta)Q \leq \hat{Q} \leq (1 + \vartheta)Q$ compare their cone sections by radial factors $(1 \pm \vartheta)^{-1/2}$. The root decoder is homogeneous of degree γ . Inclusion and homogeneity of diameter give the result. $\square$

3.3. The information hidden by one-sided endpoints. In the repeated regime write the retained variables as (x, z) , where $x \in \mathbb{R}_+^r$ are the visible interior variances and $z \in \mathbb{R}_+^e$ are the endpoint sums. The leading root scale erases z . Define

$$G_Q(x) = \min_{z \geq 0} (x, z)^\top Q(x, z). \quad (3.6)$$

Theorem 3.5 (The effective spectral metric). *For $Q > 0$, G_Q is a continuous, positive, homogeneous function of degree two on $\mathbb{R}_+^r \setminus \{0\}$, and is quadratic on each of finitely many closed polyhedral cones covering $\mathbb{R}_+^r$. More explicitly, for a subset A of the endpoint indices put*

$$z_A = -Q_{AA}^{-1} Q_{Ax} x, \quad z_{A^c} = 0.$$

On the cone defined by $x \geq 0$, $z_A \geq 0$, and

$$Q_{bx} x + Q_{bA} z_A \geq 0 \quad (b \notin A),$$

its value is

$$G_Q(x) = x^\top (Q_{xx} - Q_{xA} Q_{AA}^{-1} Q_{Ax}) x. \quad (3.7)$$

The empty subset has the evident convention. The complete leading root fibre with the fixed cluster decoder determines G_Q , and conversely. Hence two matrices yield the same leading spectrum if and only if their effective functions agree on $\mathbb{R}_+^r$; equality of the full matrices is unnecessary.

Proof. Strict convexity and coercivity in z give a unique minimizer. At a minimizer its positive support A satisfies the displayed stationarity and complementary inequalities. Conversely those conditions are the first-order necessary and sufficient conditions for this convex quadratic program. The inverse of the principal positive definite block exists, and substitution gives (3.7). There are finitely many subsets; their conditions are linear homogeneous inequalities in x and cover the cone, including boundaries where several descriptions agree. Continuity follows either from this finite agreeing description or from uniqueness and coercivity of the minimizer. Positive definiteness gives $G_Q(x) \geq \lambda |x|^2$ for some $\lambda > 0$, and scaling the minimizer proves homogeneity.

The spectrum recovers the set $\{x \geq 0 : G_Q(x) \leq 4\}$ by reading the variances of its repeated root blocks. Every variance vector is realizable by ordered zero-sum root blocks. Its radial function recovers G_Q by homogeneity and positivity. Conversely (3.6) and the decoder recover precisely the leading root fibre. This proves both implications. $\square$

Example 3.6. With one visible variance and one endpoint sum, let $Q_b = \begin{pmatrix} 1 & b \\ b & 1 \end{pmatrix}$, $0 \leq b < 1$. Then $G_{Q_b}(x) = x^2$ for $x \geq 0$, although the matrices and their mixed ray costs vary with b . For $-1 < b < 0$, instead $G_{Q_b}(x) = (1 - b^2)x^2$. The finite active-face formula detects this difference. It also explains why a full-matrix lower bound cannot be asserted for every endpoint-decoded fibre.

4. POLYNOMIAL IDENTIFICATION WALLS THROUGH CANCELLATION

Fix integers $1 \leq m < n_1 < \dots < n_s$ and let $a = (a_1, \dots, a_s)$ range over a compact subset of $\mathbb{R}^s$. On a sufficiently small fixed interval $0 \leq x \leq \epsilon$, put

$$g_a(x) = \sum_{i=1}^s a_i x^{n_i}, \quad R_a(x, \delta) = (x^m - \delta, g_a(x)).$$

For $H = (h_{ij}) > 0$, define $d_H(a, \delta) = \min_x \|R_a(x, \delta)\|_H$. Write

$$z = \delta^{1/m}, \quad C(a, z) = g_a(z), \quad \eta = n_1 - m > 0, \quad \kappa_H = (h_{22} - h_{12}^2/h_{11})^{1/2}. \quad (4.1)$$

The scalar C is a polynomial in the ramified time and the model coefficients. It is not a truncated minimum-distance expansion.

Theorem 4.1 (Cancellation-uniform wall factorization). *For every compact coefficient set and compact set of positive definite H , there are $L < \infty$ and $z_0 > 0$ such that for $0 < z < z_0$,*

$$(1 - Lz^\eta)\kappa_H|C(a, z)| \leq d_H(a, z^m) \leq (1 + Lz^\eta)\kappa_H|C(a, z)|. \quad (4.2)$$

In particular $d_H(a, z^m) = 0$ if and only if $C(a, z) = 0$. The estimate is uniform through that exact wall, including coefficient paths on which arbitrarily many leading terms cancel.

For a fixed nonzero coefficient vector, let j be its first nonzero index. Then the entrance exponent and coefficient are n_j/m and $\kappa_H|a_j|$. More generally, along any bounded semialgebraic coefficient path $a(\delta)$, either $C(a(\delta), \delta^{1/m})$ is eventually zero, giving an exact remote representative, or

$$C(a(\delta), \delta^{1/m}) = c\delta^\rho + o(\delta^\rho), \quad c \neq 0, \quad d_H(a(\delta), \delta) = \kappa_H|c|\delta^\rho + o(\delta^\rho). \quad (4.3)$$

Here $\rho \geq n_1/m$. Thus this single finite polynomial classifies every arc opening in the family, not just the generic opening on each fixed-coefficient stratum.

Proof. All constants below are uniform on the compact parameter sets. The trial $x = z$ gives $d_H \leq C_1|C(a, z)|$, and $|C(a, z)| \leq C_2 z^{n_1}$. At any minimizer, norm equivalence therefore gives

$$|x^m - z^m| \leq C_3|C(a, z)| \leq C_4 z^{n_1}.$$

For small z , x/z lies in a fixed neighbourhood of one and $x/z = 1 + O(z^\eta)$. Set $U = x^m - z^m$. The exact divided-difference identity is

$$g_a(x) = C(a, z) + UB_a(x, z), \quad B_a(x, z) = \sum_i a_i \frac{x^{n_i} - z^{n_i}}{x^m - z^m}. \quad (4.4)$$

At $x = z$ use the continuous value $\sum_i a_i(n_i/m)z^{n_i-m}$. For x/z near one, factoring z in each quotient shows $|B_a(x, z)| \leq C_5 z^\eta$. Thus the shear $(U, C) \mapsto (U, C + B_a U)$ changes the H -norm by a relative factor between $1 - C_6 z^\eta$ and $1 + C_6 z^\eta$. This is a uniform operator-norm estimate; it does not divide by C .

Completing the square gives

$$\min_{U \in \mathbb{R}} \|(U, C)\|_H = \kappa_H|C|, \quad U_* = -h_{12}C/h_{11}.$$

Applying the lower shear estimate at a minimizer proves the lower bound in (4.2). For the upper bound take $x_* = (z^m + U_*)^{1/m}$. Since $U_* = O(z^{n_1})$ and $n_1 > m$, this is positive, lies in the fixed interval, and has $x_*/z = 1 + O(z^\eta)$. The same divided-difference and shear estimates apply,

proving the upper bound. If $C = 0$, the trial $x = z$ is exactly a zero of the residual. Conversely a zero residual must have $x = z$ and hence $C = 0$.

For fixed coefficients the first nonzero monomial of C gives the stated order and coefficient. For a bounded semialgebraic path, the substituted scalar is semialgebraic in positive time. It is eventually zero or has a nonzero Puiseux leading term. Boundedness of the coefficients gives $|C| = O(\delta^{n_1/m})$, hence the asserted lower bound on ρ . The uniform relative estimate, rather than an absolute remainder at a generic stratum, proves (4.3) even after cancellation. $\square$

Corollary 4.2 (The entire metric envelope on an opening arc). *On a nonzero arc in (4.3), the leading residual set in the two displayed residual coordinates is*

$$\mathcal{E}_\rho = \{(u, c) : u \in \mathbb{R}\}.$$

Consequently the canonical metric envelope is

$$\text{cl}(\text{conv}\{(u, c)(u, c)^\top : u \in \mathbb{R}\} + \text{Sym}_2^+),$$

and its support data are exactly $a_H^2 = c^2(h_{22} - h_{12}^2/h_{11})$.

Proof. A bounded normalized residual forces $U = O(\delta^\rho)$ and hence $x/z \rightarrow 1$, since $\rho > 1$. Equation (4.4) and $B_a = O(z^\eta)$ make its second normalized coordinate tend to c . Conversely every fixed u is realized by $x = (\delta + u\delta^\rho)^{1/m}$, which is feasible for small positive time. The same identity gives its normalized residual limit (u, c) . The last assertion follows by completing the square, or by Theorem 2.2. $\square$

Corollary 4.3 (Crossover and higher cancellation). *Let $s = 2$, $n_1 = n < n_2 = q$, and fix the second coefficient b . For $a_1 = A\delta^{(q-n)/m}$,*

$$\delta^{-q/m} d_H(a, \delta) \longrightarrow \kappa_H |A + b|$$

uniformly for A in compact sets, including $A = -b$. At $A = -b$ the residual vanishes exactly for all small positive time. If instead

$$a_1 = -b\delta^{(q-n)/m} + c\delta^\lambda, \quad \lambda > (q-n)/m, \quad c \neq 0,$$

then the opening order is $n/m + \lambda$ and its coefficient is $\kappa_H |c|$.

Proof. Substitution in the exact polynomial gives respectively $C = (A + b)\delta^{q/m}$ and $C = c\delta^{n/m+\lambda}$. Apply (4.2). Uniformity includes the zero coefficient because the estimate is relative to $|C|$, with both sides zero on the wall. $\square$

Corollary 4.4 (What valuations and residues determine). *Suppose the nonzero coefficient germs have leading terms $a_i(\delta) = b_i\delta^{\lambda_i} + o(\delta^{\lambda_i})$. Put $r = \min_i(\lambda_i + n_i/m)$. If*

$$\sum_{i: \lambda_i + n_i/m = r} b_i \neq 0,$$

then the opening has order r and coefficient κ_H times the absolute value of this sum. If the sum vanishes, neither the order nor exact membership in the wall is determined by these leading valuations and residues alone; one must evaluate the next nonzero term of the exact scalar C , or determine that it vanishes identically.

Proof. The first assertion is the leading-term rule for the finite sum defining C , followed by Theorem 4.1. For the second, fix $b \neq 0$ in Corollary 4.3. The paths $a_1 = -b\delta^{(q-n)/m}$ and $a_1 = -b\delta^{(q-n)/m} + c\delta^\lambda$ have the same leading valuations and residues, but the first is exact and the second has opening order $n/m + \lambda$. Different $\lambda > (q-n)/m$ give different nonexact orders with the same initial data. $\square$

This does not claim that a fixed finite truncation of an arbitrary parameter path suffices. The finite invariant is the exact model polynomial C ; a fixed-format path has the finite order bounds supplied by the accompanying semialgebraic theory. In the language of real weighted comparison, the residue sum is the obstruction to lifting the putative lowest-order opening, and the cancellation locus is an actual feasible identification wall.

4.1. Strictly positive probability observations. Choose a strictly positive probability vector P_0 with at least three cells and linearly independent zero-sum vectors A, B . On two disjoint sufficiently short closed intervals define

$$P_{\text{base}}(s) = P_0 + As, \quad P_{\text{rem},a}(x) = P_0 + Ax^m + Bg_a(x), \quad s, x \geq 0. \quad (4.5)$$

The interval lengths may be chosen uniformly for bounded a so that all probabilities stay strictly positive. Give the two components distinct constant monic degree-one target polynomials. The exact parameter fibre over P_0 consists of the two origins, even when some or all coefficients vanish. Use the centre path $s = \delta$.

Theorem 4.5 (Hellinger wall with its sharp coefficient). *For the polynomial probability model (4.5), let $d_h(a, \delta)$ be the minimum Hellinger distance from the centre to the remote component. With*

$$\langle Y, Z \rangle_I = \sum_e Y_e Z_e / P_{0,e}, \quad \kappa = \frac{1}{2} \left(\|B\|_I^2 - \frac{\langle A, B \rangle_I^2}{\|A\|_I^2} \right)^{1/2} > 0,$$

there are uniform constants such that

$$(1 - Lz^{\eta_0})\kappa|C(a, z)| \leq d_h(a, z^m) \leq (1 + Lz^{\eta_0})\kappa|C(a, z)|, \quad \eta_0 = \min\{m, n_1 - m\} > 0. \quad (4.6)$$

All exact walls, arc openings and crossover laws of Theorem 4.1 and its corollaries hold with this coefficient. In particular the family realizes each of its successive rational orders and their moving-parameter cancellations while preserving strict positivity and a two-point central exact fibre.

Proof. Let $U = x^m - \delta$ and $V = g_a(x)$. The probability residual is $AU + BV$. Independence and positivity make its Hellinger norm uniformly equivalent to the Euclidean norm of (U, V) on the fixed compact neighbourhood. The trial $x = z$ and the lower norm bound localize every minimizing x exactly as in Theorem 4.1. Thus $x^m = \delta + O(|C|)$, $g_a(x) = O(|C|)$, and both probability vectors are $P_0 + O(\delta)$.

Use the exact identity

$$h(P', P)^2 = \sum_e \frac{(P'_e - P_e)^2}{(\sqrt{P'_e} + \sqrt{P_e})^2}.$$

At the localized points its denominator weights equal $(4P_{0,e})^{-1}(1 + O(\delta))$, uniformly. Hence Hellinger distance is the norm induced on (U, V) by

$$H_0 = \frac{1}{4} \begin{pmatrix} \|A\|_I^2 & \langle A, B \rangle_I \\ \langle A, B \rangle_I & \|B\|_I^2 \end{pmatrix}$$

times a relative factor $1 + O(\delta)$. The localization and divided-difference proof apply to the Hellinger minimizer for the lower bound and to the quadratic minimizing trial for the upper bound. Combining $O(z^{n_1-m})$ and $O(z^m)$ gives (4.6). The Schur complement of H_0 is κ^2 . Finally independence of A, B makes $P_{\text{rem},a}(x) = P_0$ imply $x^m = 0$, and the base equation implies $s = 0$. The exact central fibre and positivity assertions follow. $\square$

Remark 4.6 (Closed observation balls and critical thresholds). The remote component enters a closed observation ball of radius $t(\delta)$ exactly when $d_h(a, \delta) \leq t(\delta)$. The factorization determines this decision when the limiting ratio is different from one, and it identifies the exact wall $C = 0$

without an asymptotic test. When a nonzero threshold and distance have the same leading term, the relative estimate does not assign a side. The accompanying exact semialgebraic sign test for $t - d_h$, or the signed higher-order calculations in the regular cubic cases, remain necessary. Equality is inclusion by compact attainment.

APPENDIX A. REAL MONOMIAL PRESENTATIONS AND THE CLASSICAL ORDER FORMULA

For comparison with a divisorial calculation, suppose the residual is Nash near its feasible set. A finite real monomial presentation in this appendix means a finite disjoint union of compact semialgebraic corner sectors Y_α and a proper surjective semialgebraic map

$$\pi : \coprod_{\alpha} Y_{\alpha} \longrightarrow \{(x, \delta) : x \in N, 0 \leq \delta \leq \delta_1\}$$

onto the germ under consideration, after discarding components with identically zero time and restricting to the closure of the positive-time part. On each sector, in its own dimension, the pullbacks have the form

$$\delta \circ \pi = u(r, z)r^a, \quad R \circ \pi = r^b V(r, z), \quad r_i \geq 0, \quad (\text{A.1})$$

where u is positive, V has no common zero, and these units have positive lower bounds on the compact sectors. A face is used only when it has a relative open part accessible from positive-time feasible points by an arc transverse to that face with the other boundary coordinates positive. All face indices below refer to such sectors and faces. This definition specifies the properness, image, and accessibility used by the calculation; it does not identify every real point of a complex exceptional divisor with a feasible direction.

Simultaneous principalization and real rectilinearization of a *finite* list of ideals and inequalities supply presentations of this kind in the setting treated in the supporting volume. The calculation below is also valid whenever a presentation with these properties is supplied directly. In particular its conclusions do not require an assertion that a single modification monomializes every polynomial in an infinite ring.

Proposition A.1 (Computing the intrinsic pair). *In a presentation (A.1) satisfying the nonexact hypothesis, every accessible index with $a_i = 0$ has $b_i = 0$. The exponent of Theorem 2.2 is*

$$\rho = \max_{\alpha, i: a_i > 0} b_i / a_i.$$

For each $H > 0$, the function

$$\frac{\delta^\rho}{\|R\|_H} \circ \pi = \frac{u^\rho}{\|V\|_H} \prod_i r_i^{\rho a_i - b_i}$$

extends continuously to the zero-time preimage. If M_H is its maximum there, then $0 < M_H < \infty$ and $a_H = M_H^{-1}$. Thus these resolved unit maxima reconstruct the same canonical envelope by (2.4), even when the lists of coordinate divisors in two presentations differ.

Proof. If $a_i = 0 < b_i$, the relative open face with the other coordinates positive maps to a feasible positive-time zero residual, contradicting (2.1). With ρ equal to the displayed maximum, every exponent $\rho a_i - b_i$ is nonnegative. The unit ratio is therefore bounded and continuous. On a transverse arc at a maximizing face it has a positive limit; the image of that face has zero time and zero residual, hence is the unique central solution. It follows that its zero-time maximum is positive and finite.

On each positive time level, surjectivity and compactness give

$$\max_{\delta \circ \pi = \delta} \frac{\delta^\rho}{\|R\|_H} \circ \pi = \frac{\delta^\rho}{d_H(\delta)}.$$

A sequence of maximizers has a convergent subsequence in the compact source, giving the upper limit M_H . Conversely curve selection at a zero-time maximizing point gives an accessible arc on which the unit ratio tends to M_H . Its positive time coordinate is, after restriction, strictly monotone and covers every sufficiently small positive time. This gives the reverse limit and $d_H \sim \delta^\rho/M_H$. It identifies the exponent with the intrinsic one, as does the weighted-average formula for the orders of a lifted arc. Finally Theorem 2.2 reconstructs the envelope from these metric coefficients. $\square$

The exponent in this proposition is the classical relative real contact or Łojasiewicz exponent. The normalized-blow-up and integral-closure theory explains why divisorial orders are the correct algebraic mechanism [1]. Ideals with equal integral closure have equal orders on every arc: an integral equation would otherwise have a unique term of least order. This explains order invariance, but not metric invariance. Multiplying a residual vector by a positive constant leaves its ideal and contact exponent unchanged, while scaling all entrance coefficients and its tensor envelope. Likewise, changing feasible real sectors can remove arcs without changing the ambient complex ideal. We therefore use the intrinsic arc supremum and tensor envelope as the statement of the invariant. No canonical minimal list of real-accessible Rees valuations is asserted for arbitrary inequality-constrained germs.

APPENDIX B. PRESERVATION, SCOPE, AND DEPENDENCE OF THE RESULTS

The present article has three proof dependencies. The canonical metric theorem needs compact semialgebraic minimization and one-variable real parametrization. The local ray-probe theorem needs the binary coefficient inverse and the explicit root-cluster expansion. The polynomial wall factorization needs only the divided-difference identity, localization of minimizers, and a positive definite metric. Thus the last two do not depend on the existence of a general resolution atlas.

The complete preceding article remains the supporting volume. Under the hypotheses and notation of its cubic-fibre theorem, the identifying locus for the rank-deficient cubic family $f = z(z - r)^2$, $g = (z - s)^3$ is

$$\{\det U \neq 0\} \cup \{(3D - r)s - 2rD < 0\} \cup \{(1 - \alpha)(r + 3s)(2r - 3s)^2 - 4\pi r^3 < 0\}.$$

Its direct fibre proof, inverse through rank loss, whole-model local analysis, and both wall calculations are unchanged. The new probability family in Section 4 belongs to the general polynomial observation class used in the accompanying singular-entrance theorem; it is not claimed to be a reparametrization of this cubic binary subfamily. In particular the singular rational openings do not replace the regular cubic order-one calculations.

The original finite constrained-value theorem remains valid with its supplied marking. Section 3 identifies that marking from the exact centre law and gives its costs a precise law-space oracle interpretation. The new lower bound concerns the declared variable-metric class; no sample complexity, operational intervention, or fixed-Hellinger minimality is inferred from it. The endpoint quotient in Theorem 3.5 retains the information actually seen by the spectrum rather than claiming an injective decoder where there is none.

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